Pedestrian Tracking in Dynamic Urban Scenes Using YOLO11x and OC-SORT with Camera-Motion Compensation

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Abstract*—Reliable automatic detection and tracking of pedestrians is important for traffic safety and autonomous driving, yet remains challenging in dynamic urban scenes. Clustered backgrounds, occlusions, pedestrians moving in different directions at varying distances, and continuous ego-motion of the camera often lead to fragmented trajectories and identity switches, which degrade the system's ability to anticipate pedestrian intent. This paper investigates and compares several tracking strategies, ranging from Kalman-filter-based single-object trackers to modern multiple-object tracking approaches such as Norfair, DeepSORT, and OC-SORT, and proposes enhancements that compensate for camera motion and improve tracking through occlusions. The proposed modifications reduce identity switches and extend the duration of consistent pedestrian tracks. Experiments on the MOT17 benchmark demonstrate that the best performance, with MOTA of 40.4% and IDF1 of 51.7%, is achieved by a configuration that combines a YOLO11x detector with the OC-SORT tracker augmented by ByteTrack association logic and a camera-motion compensation (CMC) module. These results indicate that reliable perception for autonomous driving requires coupling robust object detection with explicit camera-motion modelling and advanced data-association techniques within the MOT pipeline.

***Keywords*—computer vision; person detection; person tracking; YOLO architecture; OC-SORT algorithm; ByteTrack algorithm; CMC algorithm**

I. INTRODUCTION

Multiple Object Tracking (MOT) is a fundamental component of modern perception systems, particularly in autonomous driving and intelligent transportation, where reliable and temporally consistent tracking of pedestrians is essential for safe navigation and decision making. Unlike single-frame object detection, MOT must maintain object identities over time under challenging real-world conditions such as frequent occlusions, illumination changes, viewpoint variations, and camera motion, which are especially pronounced in dense urban environments. In such scenes, pedestrians exhibit highly variable and often unpredictable motion patterns, including sudden changes in direction, varying speeds, and complex interactions with other agents, which substantially increase tracking complexity and the risk of safety-critical situations [1]. Consequently, accurate detection combined with robust long-term tracking of pedestrians is a key requirement for collision avoidance, motion planning, and higher-level scene understanding and intent prediction in autonomous vehicles [2].

Recent advances in deep learning have significantly improved person and pedestrian detection performance, even under adverse conditions with cluttered backgrounds, limited visibility [3], and dynamic scenes [4]. However, in real-world traffic scenarios, tracking quality is still strongly affected by frequent occlusions, dense crowds, strong perspective effects [5], and the continuous ego-motion of the camera, which induces apparent background motion that complicates motion modelling and data association [6]. These factors often lead to fragmented trajectories and identity switches, degrading the reliability of downstream perception and decision-making modules in autonomous driving stacks [7].

Traditional tracking approaches based on linear motion models and Kalman filtering provide computational efficiency, but typically fail in crowded scenes and under long occlusions [7]. That motivated tracking-by-detection pipelines that combine deep neural detectors with more advanced data-association strategies [8]. Some of modern MOT methods are Norfair [16], DeepSORT [15], ByteTrack, [10] and OC-SORT (Observation-Centric SORT) [11]. By design, their performance depends largely on the quality of the detector, the association logic and the proper selection of hyper parameters. In addition, several studies have shown that even state-of-the-art tracking methods show significant performance degradation when camera motion is not explicitly modeled [6], [7], [9].

The goal of this paper is to systematically evaluate and enhance pedestrian tracking performance in dynamic urban environments using MOTChallenge benchmark [14] that provides diverse urban video sequences with annotated pedestrian trajectories and a common evaluation protocol, making it a widely adopted reference for pedestrian MOT research.

We investigate several tracking configurations that combine a YOLOv11 detector with modern trackers and alternative association strategies, and we introduce a camera-motion compensation module designed to better handle ego-motion-induced background changes. In particular, we study how design choices such as detector confidence thresholds, input resolution, data-association logic, interpolation of missing detections, and explicit camera-motion compensation affect standard MOT metrics, with a focus on improving MOTA and IDF1.

The remainder of the paper is organized as follows: Section II details the proposed tracking pipeline, including the YOLO11x detector, OC-SORT framework, and camera-motion compensation. Section III presents the experimental setup, MOT17 dataset configuration, and quantitative results. Section IV presents and discussed the experimental results and Section V concludes the paper and outlines directions for future research.

II. Proposed Tracking Algorithm

We examined different configurations of pedestrian tracking method based on automatic detection and tracking algorithm and proposed a system that integrates YOLO11x object detector with improved OC-SORT tracker.

OC-SORT is observation-oriented and to improve tracking performance in challenging dynamic urban scenes, we improved the detection association strategy and added interpolation and camera motion compensation to reduce camera drift during recording.

The tracking core is based on the OC-SORT algorithm, which we extended with three components:

1. **Camera Motion Compensation (CMC),**
2. **ByteTrack Association Logic,**
3. **Trajectory Interpolation.**

We implemented the CMC module using the Lucas-Kanade optical flow method [12] to separate ego-motion (e.g. dashboard cameras) from pedestrian motion. Features are tracked between consecutive frames to estimate an affine transformation matrix (translation, rotation, and scaling). This transformation is applied to the Kalman filter state vector (position and velocity) before the prediction step, ensuring that the motion model takes into account the camera motion.

To mitigate the fragmented trajectories caused by occlusion, we adopt a two-stage data association strategy from the ByteTrack algorithm [10]. High-confidence detections ($score \geq 0.5$) are first associated using Intersection over Union (IoU). The remaining low-confidence detections ($0.05 \leq score < 0.5$) are then compared to the mismatched tracks [10]. This allows tracking of partially occluded or blurred objects, which typically show lower detection scores.

Additionally, linear trajectory interpolation was applied post-processing to fill gaps in trajectories up to 20 frames long, which significantly improved the recall of the system.

The proposed hybrid algorithm is further referred to in the text as: Y11x + OC-SORT + BT + CMC.

III. Experimental Setup

The experiment was designed with incremental changes to gain insight into which system components and mechanisms provide the most significant improvements under realistic constraints. First, the baseline tracking model configurations were tested, combining different detectors and tracking algorithms and then individual tracking components were tested to isolate their contribution to performance.

A. *Detection and tracking methods*

For the detector, we used l and x model variants from the YOLOv11 family [13], both previously trained on the COCO dataset without additional fine-tuning.

Then, we tested detection models as an integral component of selected modern tracking algorithms: Norfair, DeepSORT, OC-SORT, and ByteTrack and then on the best performing

model we examined how various parameters such as detector confidence thresholds, input image resolution, associated detection association strategies, and camera motion interpolation and compensation affect the overall tracking performance

B. *Evaluation dataset*

We performed the evaluation on the MOT17 benchmark, a part of MOTChallenge benchmark [14], which includes diverse urban scenes captured by a surveillance camera and a motion camera with pedestrians and vehicles in motion, moving individually and in groups, entering and exiting the camera's field of view from different directions, at different distances from the camera. An example of images from the MOT17 dataset is shown in Fig 1. The used MOT17 dataset contains 14 video sequences, two of which were taken from the test set and the remaining were used for training.

Figure 1. An example of images from the MOT17 dataset

Video sequences in MOT17 are annotated with pedestrian trajectories and accompanied by ground truth detections at the bounding box level from multiple detectors. To adapt the MOT17 dataset for YOLO, we converted the ground truth annotations from the standard MOT format *[frame, id, left, top, width, height]* to the YOLO format *[class, x_center, y_center, width, height]*, normalized by image dimensions. Only detection belonging to the *Pedestrian* class (Class ID 1 in MOT17) were used in experiments.

We set all images to an input image resolution of 1280 x 1280 pixels.

C. *Evaluation metrics*

The existing ground truth enables tacking evaluation using standard MOTA metrics including MOTA, MOTP, IDF1, IDP, IDR, precision, recall, and IDSW, that assess both detection quality and identity preservation. A metrics brief description is given in Table 1.

TABLE I. STANDARD MOTA METRICS WITH DESCRIPTION

Metric	Description
MOTA	Multiple Object Tracking Accuracy - overall accuracy metric that takes into account FP, FN, and IDs.
MOTP	Multiple Object Tracking Precision - average overlap between the predicted frames and the ground truth.
Precision	Percentage of detected objects
Recall	Percentage of ground truth objects that are correctly detected
IDF1	Measures how consistently a tracker tracks the same object
ID Switches	Number of identity switches, number of times the tracker incorrectly changes the object ID (less is better)
MT/ ML	Mostly Tracked trajectories (higher is better) and Mostly Lost trajectories (less is better)

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. *Comparison of tracker performance with YOLOv11 detector*

Based on preliminary evaluation on the MOT17 benchmark, where all models (Norfair, DeepSORT, OC-SORT and ByteTrack) used YOLOv11 detectors (l and x variants), key trade-offs between trackers were identified (Table 2).

TABLE II. RESULTS OF TRACKING METHODS WITH YOLO11-L AND YOLO11-X DETECTORS ON THE MOT17 PEDESTRIAN CLASS

Method	Prec	Rcll	IDF1	ID Sw	MOTA	MOTP
YOLOv11- l detector						
+ Norfair	86.2%	25.9%	30.3%	369	21.4%	17.9%
+ OC-SORT	**93.2%**	24.3%	31.6%	213	22.3%	17.3%
+ DeepSORT	60%	**45.7%**	35.1%	636	14.5%	**22.5%**
+ ByteTrack	90.7%	38.8%	**45.4%**	**204**	**34.6%**	19.1%
YOLOv11-x detector						
+ Norfair	86.5%	26.0%	30.1%	345	21.5%	17.4%
+ OC-SORT	**93%**	24.5%	32.4%	**177**	22.5%	16.8%
+DeepSORT	57.5%	**49.5%**	37.4%	573	12.3%	**22.5%**
+ ByteTrack	83.6%	47.2%	**46.9%**	462	**37.4%**	20.5%

ByteTrack achieves the highest MOTA values (34.6% for YOLOv11-l and 37.4% for YOLOv11-x), thanks to superior utilization of low-score detections and minimal ID switches (204 for l variant). However, its Recall (38.8–47.2%) remains limited in occlusion-heavy scenarios. Conversely, DeepSORT has solid Recall (45.7% l, 49.5% x) and IDF1, but poor MOTA due to high ID Sw (636/573), indicating instability of identification in crowds. On other hand, OC-SORT demonstrates robustness in identity preservation, with the lowest ID switches

(177 for YOLOv11-x) and high Precision (>93%), albeit at the cost of lower Recall (~24–25%). This stability makes it promising solution for enhancing long-term tracking consistency and in crowds at MOT17.

B. *Performan evaluation of our Y11x + OC-SORT + BT + CMC algorithm*

Given the demonstrated performance (Table 2), the OC-SORT method with the YOLOv11-x detector, enhanced with the BayTrack algorithm, was selected for further experimentation to exploit the synergistic effect of ByteTrack's successful object association and OC-SORT's conservative state management with the aim of increasing IDF1 and MOTA while maintaining low ID Sw in complex MOT17 sequences. We additionally implemented camera motion compensation (CMC) algorithm that effectively aligns the frames, allowing the Kalman filter to distinguish global camera motion from local object states and post-processing steps to recover lost tracks (linear interpolation - LT).

In all cases, the detecting precision, IOU, for the YOLOv11-x detector is set to <30. The results are summarized in Table III.

TABLE III. TRACKING PERFORMANCE OF PROPOSED ALGORITHM ON MOT17 DATASET

Metric	YOLO11x + OC-SORT	YOLO11x+ ByteTrack	Y11x + OC-SORT + BT+CMC (ours)
MOTA	22.5%	37.4%	**40.4%**
IDF1	32.4%	46.9%	**51.7%**
Recall	24.5%	47.2%	**51.4%**
Precision	**93%**	83.6%	82.7%
ID Switches	**177**	462	276
MT	63	99	**138**
ML	318	162	**141**

The proposed hybrid algorithm (Y11x + OC-SORT + BT + CMC) achieved our goal of increasing the MOTA metrics (IDF1 +4,8, MOTA +3% over ByteTrack), and maintaining a low ID Sw (-188 than ByteTrack-x, +97 than OC-SORT), with significant increase in mostly tracked trajectories and decrease in lost trajectories (+39 MT, -21 ML over ByteTrack and +75 MT, -177 over OC-SORT).

C. *Impact of detector confidence on tacking performance of proposed algorithm*

Given that our proposed tracking algorithm relies heavily on the quality of observations, we separately evaluated the performance of the detector, and then the influence of confidence on the tacking results.

The performance of YOLOv11x detector is showed on precision-recall curve (Fig. 2) with mAP@0.5 of 0.8, which is promising results for the tracking algorithm, taking into account that there are frequent occlusions in the crowd and a large number of distant pedestrians occupying a small area of the image on the MOT17 set.

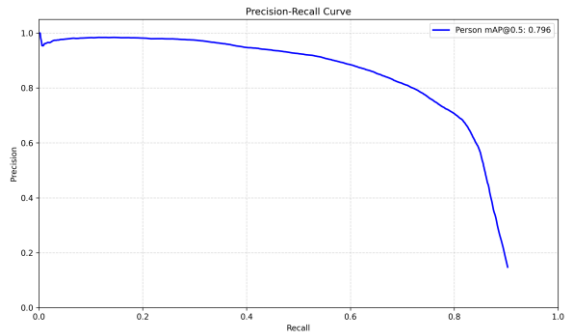

Figure 2: Precision-Recall Curve for YOLOv11x on MOTA17 test set

We also analyzed the F1-score at different confidence thresholds (Fig. 3) and it was shown that although the best F1 results are achieved with a confidence of 0.5, there is also a range from 0.1 to 0.6 in which the results are stable.

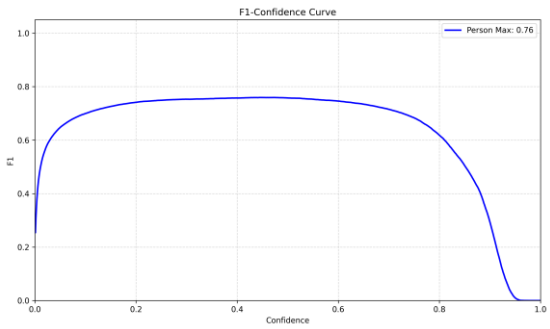

Figure 3: F1-Confidence Curve with respect to confidence threshold

We used this information to determine the most optimal confidence threshold for our tracking algorithm, in such a way that a threshold of 0.5 was used for the primary detection and a lower level of confidence was used for the following association stages. The influence of confidence thresholds on the performance of the detector (denoted as conf) and the tracker

(denoted as lo_conf) are shown in Table IV. The best values are marked in bold.

TABLE IV. COMPARISON OF TRACKER METRIC WITH DIFFERENT CONFIDENCE THRESHOLDS

Metric	conf > 0.5; lo_conf >0.05	conf > 0.5; lo_conf >0.1	conf > 0.6; lo_conf >0.1
MOTA	**40.4%**	**40.4%**	39.4%
IDF1	**51.7%**	**51.7%**	50.5%
Recall	**51.4%**	**51.4%**	47.8%
Precision	82.7%	82.7%	**85.4%**
ID Switches	276	276	**231**
MT	**138**	**138**	120
ML	**141**	**141**	186

It was shown that lowering the confidence threshold for the association did not affect the performance, however increasing the confidence threshold for +0.1 for the detector, affected the MOTA metrics, in such a way that it slightly reduced IDF1, MOTA (~1%), at the expense of decreasing ID shift (-45) but with worsen result on trajectories, which shows that the confidence should be carefully modeled considering the type of problem and the environment in which the tracking algorithm is applied.

D. *Impact of input image resolution on tacking performance of proposed algorithm*

In the next experiment, we wanted to check whether increasing the resolution of the input image (from 1280 x 1280 to 1920 x 1920 pixels) would improve the results, especially since there are a large number of smaller or distant pedestrians in the set (Table V).

TABLE V. COMPARISON OF TRACKER METRICS WITH DIFFERENT RESOLUTION DETECTORS (1920PX VS. 1280PX)

Metric	1920px	1280px	Delta
MOTA	**40.7%**	40.4%	+0.3%
IDF1	**52.1%**	51.7%	+0.4%
Recall	50.5%	51.4%	-0.9%
Precision	**84.2%**	82,7%	+1.5%
ID Switches	306	**276**	+30
Speed (in *HZ*)	**2.91**	5.14	-2.23

Increasing the resolution to 1920px resulted in a marginal increase in MOTA and IDF1 matrices, but with an increase in ID switches and slower inference, so we concluded that a resolution of 1280px is more appropriate for the practical application of our algorithm in real time.

E. *Qualitative Evaluation on Video Sequences*

Here we provide a qualitative analysis of the proposed tracking method (Y11x + OC-SORT + BT + CMC) on two videos with different challenges, MOT17-13 has fast camera movement from a moving vehicle and MOT17-02 has large occlusions in pedestrian zone.

Figure 4, shows the situation with camera in a moving vehicle and the integration of the CMC module in algorithm was important to reduce the pedestrian ID switch. Additional challenges arise during fast turns or when large vehicles block the field of view because a sudden change in the optical flow vector can temporarily disrupt the alignment of the bounding box. Also, pedestrians at the far end of the street (occupying a small image area) occasionally flicker or are assigned new IDs when occluded, indicating that while YOLO11x is robust, extreme scale variations combined with occlusion remain challenging.

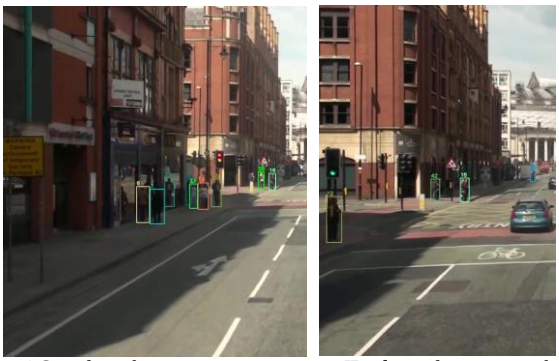

Figure 4 Our algorithm maintains consistent IDs for pedestrians on the sidewalk despite the continuous motion of the camera (MOT17-13)

Figure 5 gives an example of a pedestrian zone with a high density of pedestrians moving in different directions and often in occlusion several frames, recorded with a fixed camera. In this case, to preserve the pedestrian ID, the momentum of OC-SORT, the ByteTrack connection logic with subsequent interpolation was important to bridge the gaps where the detection results currently fall below the confidence threshold. The algorithm still struggles with longer occlusion, around +5s and when the subject is completely occluded for more than 20 frames (interpolation buffer limit), because then the track is broken and the subject is assigned a new ID. Also, in the background of

the scene where there is a large overlap of pedestrians and the bounding box are very small, fragmentation of identity occurs much more often than with subjects in the foreground. This shows that the algorithm has limitations, but that it successfully deals with situations that are of primary interest for safety and autonomous driving.

Figure 5 Our tracking algorithm on sequence MOT17-02 effectively handles dense crowd interaction, preserving ID as subjects cross paths

V. CONCLUSION

This paper evaluates state-of-the-art pedestrian tracking methods with automatic detection on the MOT17 benchmark and proposes a hybrid approach Y11x + OC-SORT + BT + CMC integrating YOLOv11x detection, OC-SORT state management, ByteTrack association logic, linear trajectory interpolation, and optical flow-based camera motion compensation. Results demonstrate that explicit camera motion modeling and multi-phase object association significantly enhance identity consistency in crowded scenes.

On the MOT17 test set, the proposed method outperforms baselines in dynamic urban environments, whether captured from moving vehicles or static cameras in pedestrian zones, achieving IDF1 51.7% and MOTA 40%, ID Sw 276.

Analysis of confidence thresholds reveals their sensitivity to task-specific scenes, necessitating adaptive tuning to optimize MOTA. Increasing input resolution beyond 1280×1280 yields only marginal gains at significantly higher computational cost, confirming 1280 as optimal for urban tracking.

Despite these advances, challenges persist in large crowds, prolonged occlusions, and distant objects. Future work will refine this hybrid algorithm via fine-tuning, incorporate re-identification modules, and evaluate on new and more demanding benchmarks like MOT20.

ACKNOWLEDGMENT

This research was supported by UNIRI project (grant:uniri-iz-25-103).

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